NuMagSANS Tutorial 5 - Longitudinal Helix Sweep

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Goal

Build a two-dimensional sweep for a longitudinal helical magnetic sphere.

- Materialize one spherical local object.
- Write 16 vector-field files `m_1.csv` to `m_16.csv`.
- Sweep the longitudinal component $\lambda = 0, \ldots, 10$.
- Write 16 orientation-distribution files `RotData_1.csv` to `RotData_16.csv`.
- Sweep $\beta_2 = 0^\circ, \ldots, 180^\circ$ while keeping $\beta_1 = 0^\circ$.

Longitudinal Helix

The local magnetic state is a transverse helix plus a constant longitudinal component.

$$
\begin{aligned}
\mathbf{m}'(z) &= \mathbf{e}_x \cos(kz + \varphi_0) + \chi\, \mathbf{e}_y \sin(kz + \varphi_0) + \lambda\, \mathbf{e}_z, \\
\mathbf{m}(z) &= \mathbf{m}'(z)/|\mathbf{m}'(z)|
\end{aligned}
$$

- `k` controls the pitch.
- `chi` controls the chirality.
- `lambda` controls the longitudinal component before normalization.
- `normalize=True` keeps the magnetization length fixed.

Sweep Parameters

Both sweeps use 16 inclusive support points.

$$\lambda_i = \frac{10}{15} i, \qquad i = 0, \ldots, 15$$
$$\beta_{2,j} = j \cdot 12^\circ, \qquad j = 0, \ldots, 15$$

- $180^\circ/(16-1) = 12^\circ$.
- λ uses 16 evenly spaced values from 0 to 10.
- $\beta_1 = 0^\circ$ for every RotData file.
- NuMagSANS later combines the 16 magnetic states with the 16 RotData files.

Python File

Create a file named `longitudinal_helix_lambda_beta_sweep.py`.

```
import math
from pathlib import Path

import numpy as np

from NuMagSANS.SystemDesigner.Workflows import (
    write_spherical_replication_vectorfield_sweep,
)
```

Field Cases

Each entry writes one file `MagData/Object_1/m_i.csv`.

```
lambda_values = np.linspace(0.0, 10.0, 16)

field_parameter_cases = [
    {
        "field_type": "longitudinal_helix",
        "k": 2.0 * math.pi / pitch_nm,
        "chirality": chirality,
        "phase": phase,
        "lambda": lambda_value,
        "normalize": True,
    }
    for lambda_value in lambda_values
]
```

RotData Cases

Each entry writes one file `RotData/RotData_j.csv`.

```
beta2_values = np.linspace(0.0, math.pi, 16)
beta1_values = np.zeros_like(beta2_values)
```

Here `beta1_values` are passed as `beta_min`, and `beta2_values` are passed as `beta_max`.

Generate Real-Space Data

The workflow writes one object, 16 magnetization files, and 16 RotData files.

```
summary = write_spherical_replication_vectorfield_sweep(
    output_dir=Path(__file__).resolve().parent / "LongitudinalHelixSweep_R20",
    R=20.0,
    a=1.0,
    atomtype="Fe",
    n_replications=1024,
    field_parameter_cases=field_parameter_cases,
    n_rotdata=len(beta2_values),
    beta_min=beta1_values,
    beta_max=beta2_values,
    rotation_seed=1,
)
```

Run Locally

Run the generator from the NuMagSANS repository root.

```
python tutorials/tutorial_5/longitudinal_helix_lambda_beta_sweep.py
```

The generated data are written to `tutorials/tutorial_5/LongitudinalHelixSweep_R20/RealSpaceData`. Set `WRITE_DIAGNOSTIC_PNG=True` in the script to additionally write an off-screen PyVista diagnostic image for each `m_i.csv`.

Generated Data

The workflow creates one real-space object and two sweep axes.

- `Local_Objects/Object_1/pos.csv`: lattice positions inside the sphere.
- `Local_Objects/Object_1/atomtype.csv`: atom labels.
- `Local_Objects/Object_1/meta.csv`: object metadata.
- `MagData/Object_1/m_1.csv` to `m_16.csv`: 16 values from $\lambda = 0$ to 10.
- `RotData/RotData_1.csv` to `RotData_16.csv`: $\beta_2 = 0^\circ, \ldots, 180^\circ$.

NuMagSANS Loop

The run script activates both sweep axes.

```
RotDataLoop_From = 1
RotDataLoop_To = 16
Loop_From = 1
Loop_To = 16
User_Selection = list(range(1, 17))
RotData_User_Selection = list(range(1, 17))
```

This creates a 16×16 simulation grid over magnetic state and orientation distribution.

Run NuMagSANS

Use the generated data as input for a NuMagSANS simulation.

```
python tutorials/tutorial_5/run_numagsans_longitudinal_helix_sweep.py
```

The script writes a config file in `tutorials/tutorial_5/LongitudinalHelixSweep_R20`, starts NuMagSANS, and stores CSV output in the dataset output folder.